

Bo Yang

B.Y.Yang@tudelft.nl — Delft, Netherlands — Personal Website

EDUCATION

Delft University of Technology, CS PhD Candidate (11/2024-Now)

Major: Computer Science(Edge AI)

Supervisor: Qing Wang, Fernando Kuipers

Field: Mobile computing (CS Rankings: #7)

Eindhoven University of Technology, MSc, Cum Laude(3%) (09/2022-08/2024)

GPA: 8.5/10

Thesis: 9/10

Major: Electrical Engineering(Signal Processing System)

Shandong University, BSc (09/2018-07/2022)

Major: Electronic Engineering

QUANTITATIVE RESEARCH PROJECTS

Polymarket Arbitrage Research & Market Data Analysis

Independent project

- Developed scripts to screen Polymarket price inconsistencies for potential arbitrage opportunities.
- Analyzed public-wallet trades, activity and closed positions using the Polymarket Data API; examined maker/taker execution and purchases of both outcomes within a market.
- Investigated BTC threshold-market arbitrage and hedging ideas, considering transaction costs, liquidity and settlement differences when assessing potential opportunities.

Insider Transaction Signal Research & Backtesting

Independent project

- Implemented a Python research pipeline combining publicly disclosed insider purchases from Finviz with daily stock prices from `yfinance`; screened purchases of at least \$100,000.
- Built a backtesting framework for 1, 3, 5, 10 and 20 trading-day holding periods; distinguished transaction dates from Form 4 disclosure timestamps and modeled a 1% entry-price premium.
- Implemented return, P&L and win-rate summaries by insider role and transaction size, with trade-level exports and visualizations for strategy comparison.

PUBLICATIONS

LightWRF: Accelerating Radio Environments Reconstruction via Redundancy Reduction in Wireless Radiance Fields, (IEEE GLOBECOM 2026)

[Paper Link](#)

I propose the idea of how to build the hybrid real- and complex-valued neural network (HNN) architecture, also developed the HNN-specific architecture search framework to reduce the dimensionality of the network into a tailored solution. I verify the viability of the HNN in comparison to a real-valued architecture with an experiment using the AudioMNIST dataset.

Hybrid Real- and Complex-valued Neural Network Architecture, (Journal on Audio, Speech, and Music Processing)

[Paper Link](#)

I propose the idea of how to build the hybrid real- and complex-valued neural network (HNN) architecture, also developed the HNN-specific architecture search framework to reduce the dimensionality of the network into a tailored solution. I verify the viability of the HNN in comparison to a real-valued architecture with an experiment using the AudioMNIST dataset.

Successive Threshold-based Multipath Mitigation aided by Neural Network for UWB Ranging, (JC&S 2024)

[Paper Link](#)

I employ IMEC's existing threshold-based multipath mitigation algorithm, known as STM, and confirm that STM enhances ranging performance in multipath environments. Additionally, I propose a framework that utilizes the Channel Impulse Response (CIR) signal for range estimation, aiming to improve performance in multipath

conditions by augmenting STM with a neural network architecture called STMnet. Through comparative analysis with other methods, I demonstrate that this approach surpasses the state-of-the-art.

Patent: A wireless intelligent sensor and its application

patent number: CN202110094598.5

WORKING EXPERIENCE

NXP: [12/2023-09/2024]

Research Intern

I am a part of the ML-AI department at NXP, developing the innovative Hybrid Complex- Real-Value Neural Network. My primary focus lies within the realm of autoML and audio, particularly in areas such as Hyperparameter Optimization (HPO), Neural Architecture Search (NAS), and speech enhancement.

IMEC: [07/2023-12/2024]

Research Intern

In my role as a research intern with the UWB4z group at IMEC, my primary responsibility is to enhance the accuracy of range estimation and localization for UWB devices in diverse environments.